

MATEO WU CHEN

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EDUCATION

Northeastern University, College of Engineering – Boston, MA

December 2026

Candidate for Bachelor of Science in Computer Engineering, Minor in Computer Science

GPA: 3.6

Relevant Courses: Analysis Random Phenomena, Digital Design & Computer Organization, Embedded Design Enabling Robotics, Fund of Linear Systems, Fund of Electronics, Fund of Networks, Communication Systems, Integrated Circuit Design

Awards and Activities: Dean's list, Generate (Member, Tech lead, and Chief), NU Robotics Club, Forge Silicon Chip Design

EXPERIENCE

SharkNinja – Needham, MA

Jan 2026 – June 2026

Electrical Engineering Coop

- Developed a wireless ESP32-C3 data logger PCB and standardized appliance-testing pipeline using BLE, UART, and timestamped microSD logging, accelerating real-time diagnostics, calibration, and validation across product platforms.
- Supported development of next-generation products by contributing to analog and digital circuit design, board bring-up, and validation of embedded systems including microcontrollers, power architectures, motors, and mixed-signal interfaces.
- Engineered a Python (PyVISA) automated test platform to remotely control oscilloscopes and signal generators via Ethernet, reducing manual test time by ~90% through automated frequency sweeps, measurement acquisition, and report generation.

Northeastern University SINRG – Boston, MA

Jan 2025 – Dec 2025

Undergraduate Research Assistant - Spatial Intelligence Research Group

- Contributed to the development of a prototype AI glasses system by designing custom 4-Layer PCB using Altium Designer, assisting in the schematic capture and circuit design, including power regulation, and microcontroller interfaces.
- Developed and validated embedded firmware, performing software and hardware integration testing using UART and I2C communication protocols to verify sensor, optics, and microcontroller functionality for real-time AI-assisted applications.

Tatum Robotics – Watertown, MA

Jan 2025 – July 2025

Embedded Systems & Hardware Engineer Coop

- Designed and validated 4-Layers custom power, button, and RGB PCBs in Altium Designer for the TatumT1 Assistive Touch Hand, significantly improving feedback functionality, and contributing to a 100% functional prototype success rate.
- Migrated legacy PWM control from RPi.GPIO to gpiod on Raspberry Pi 5, programmed and debugged Python and C# applications, integrated I2C devices, and optimized software-hardware interaction to improve system response time by 80%
- Executed electrical testing and validation of robotic prototypes using oscilloscopes and digital multimeters, achieving a 95% fault detection rate and supporting reliable system operation.

Northeastern University NUCAR – Boston, MA

May 2023 – Dec 2024

Undergraduate Research Assistant – Computer Architecture Research Group

- Developed and benchmarked parallel C++ applications using CUDA, Pthreads, and OpenMP on Northeastern's Discovery Cluster, achieving up to 1000× speedup over CPU implementations through GPU acceleration.

TECHNICAL SKILLS

Software: Altium, Bash, C, C++, Git, KiCad, LTspice, Magic Layout, MATLAB, Python, Ngspice, SystemVerilog, Vivado

Hardware: FPGA Design, Logic Analyzers, Oscilloscopes, Digital Multimeters, PCB Bring-up debugging & Soldering

PROJECTS (More in Portfolio)

Xilinx FPGA-SoC Custom PCB - *Personal*

Jan 2026 – Sept 2026

- Designed a custom 6-layer Xilinx FPGA hardware platform, applying multilayer power delivery, clocking, SPI Flash, DDR3L Memory, and high-density BGA routing with controlled-impedance, and signal considerations for high-speed digital design.

Growver Tree Planter Robot - *Generate Northeastern's Product Development Studio*

Jan 2026 – April 2026

ECE Tech Lead

- Led end-to-end electrical system design for Growver, architecting, designing, and validating custom MCU, BMS, and Power Distribution PCBs to support motor control, sensing, power management, and embedded communications for autonomous planting and remote-controlled rover operations.

Candle Maker – *Generate Northeastern's Product Development Studio*

Sept 2025 – Dec 2025

ECE Hardware Member

- Designed a custom 4-layer PCB in Altium Designer for an automated candle-making device, integrating microcontrollers, sensors, power regulation, heating control, and safety circuitry.